What is a Printed Circuit Board (PCB)?

Printed Circuit Board is the most common name of what can also be called Printed Wiring Cards or Printed Wiring Boards. Before the PCBs were invented, circuits were connected through the hard and laborious process of point-to-point wiring, which was arduous as well as problematic. The point-to-point circuits were used to encounter frequent failures at wire junctions and also short circuits when wire insulation used to age and crack.

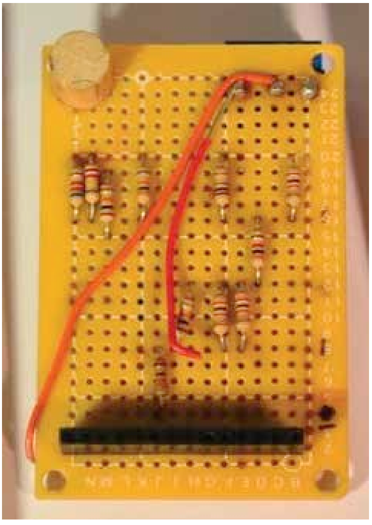
Electronics started becoming more advanced from vacuum tubes to relays to finally silicon and Integrated circuits (ICs), the size, as well as cost of production of these electronic components, started to decrease drastically and electronics became more present as consumer goods. This demand led to innovations for reducing the size and cost of production, and thus, PCBs were born.

A Printed Circuit Board is a board consisting of lines and pads that connect various points of a circuit. A PCB allows signals and power to be routed using physical devices and solder is the metal that makes connections between the surface of the PCB and electronic components which also serves as a strong mechanical adhesive.

Components of a PCB

A PCB is like a layered cake with numerous layers of different materials that are laminated together using heat as well as adhesive in a way that they work as a single object.

Now, let us look at the basic components of a Printed Circuit Board.



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1. FR4

This is the base material or substrate, which is usually made out of fiberglass. The most common designator of this fiberglass, historically has been FR4, hence the name.

If you are wondering where the rigidity of a PCB comes from, then this solid core object is responsible for the same, but nowadays there are also flexible PCBs being built